

구현 검증 보고서

구현 상태: VERIFIED · 보고서: FINAL · Science Knowledge Release: NOT ACTIVE

호출자 숫자가 아니라 Git 상태와 해시로 묶인 기계 증빙에서 구현 상태를 계산했습니다. 이 결과는 지식 Release 활성화나 과학적 진실 보증이 아닙니다.

증빙	상태	기계 결과
Suite contract	VERIFIED	schema=science-test-suite-contract/0.1, suites=3
Science JUnit	VERIFIED	tests=1689, failures=0, errors=0, skipped=0
MCP JUnit	VERIFIED	tests=23, failures=0, errors=0, skipped=0
Full regression	VERIFIED	tests=2296, failures=0, errors=0, skipped=9
Browser captures	VERIFIED	checks=27, captures=3
Equation Knowledge	VERIFIED	assets=6, catalog=6, pdf-safe=6
Qualification	VERIFIED	cases=440
Independent review	VERIFIED	status=passed, critical=0, important=0

검증 식별자

Code revision: 02f89be5caad41f45f57af7558a4b3e69c42efbc

Git dirty: False

Git status digest: sha256:e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

Evidence bundle: sha256:404f247418254fbbabca94ec124be5200a36525eacd0fb961320be558561485b

Report record: sha256:e13ff8200856d2cda1cd8a54491ed9a3bee502c073b74d03da61fe807661ed0b

상태 하향 사유

없음

Release Gate

Gate	상태	근거
G0	PASS	Schema, IDs, references, immutable component digests, and candidate lifecycle are valid.
G1	PASS	Source/Evidence hashes and locator structures are internally reproducible.
G2	PASS	Every public decision remains inside release-pinned Knowledge and Evidence.
G3	PASS	All 440 public cases match their expected verdict or ambiguity gate.
G4	PASS	Repeated public qualification is byte-stable.
G5	PENDING	Independent sealed holdout is not commissioned.
G6	PENDING	Web/REST/MCP/explanation/export parity must be established by the authoritative post-integration checker; caller assertions cannot pass this gate.
G7	PENDING	Authorized human Admin review and activation must be resolved from the operational audit store; caller assertions cannot pass this gate.

완료 경계

사람 Admin 승인과 독립 sealed holdout이 없으면 Candidate를 활성화하거나 운영 검증 완료로 표현하지 않습니다.

Candidate 수식 표시 QA

아래 6개 수식은 digest-bound Science Equation Knowledge와 정확히 일치하는 로컬 SVG를 ReportLab vector drawing으로 변환한 표시 검수 표본입니다. Release Candidate 표시이며 verdict:빨간 표시:운영 판정 권한은 없습니다.

Chemistry · deterministic_rule $c = \frac{n_{\text{solute}}}{V_{\text{solution}}}$ c = n_solute / V_solution sci:equation:chemistry:molar-concentration-definition sha256:2db62840faed18a6cfff923caf83ab6dd489648d6db0d522e5f444e1d4dcfbf8a Evidence: sci-evidence:chemistry:amount-concentration Locator: Molarity: mole/volume basis	Circuits · deterministic_rule $\sum_k V_k = 0$ sum(loop voltages with sign) = 0 V sci:equation:circuits:kvl-loop-balance sha256:ae4be4aacf85dde9c31fd93a090f0a378aeeb3470c115ca95a8a30165e8147ef Evidence: sci-evidence:circuits:kvl-law Locator: Transcript — Lecture 2, Kirchhoff laws
Materials Science · explanation_only $D = D_0 \exp\left(-\frac{E_a}{k_B T}\right)$ D = D0 * exp(-Ea / (kB*T)) sci:equation:materials:arrhenius-diffusion sha256:65cf2f1245b30cf30cb8f9c2dc39f3d5ace24360ae92fdbcfca48c825a772a99 Evidence: sci-evidence:materials:diffusion-arrhenius Locator: Recitation 26 — Arrhenius relationship	Physics · deterministic_rule $W_{\text{applied}} = \Delta K$ W_applied = delta_K sci:equation:physics:applied-work-kinetic-energy-change sha256:09ee2b7f81aa3fddd1f45e7ceb636053c7fc0f6a1b10cd57978a597e9d013bf5 Evidence: sci-evidence:physics:work-energy-power Locator: 13.6 Work-Kinetic Energy Theorem
Semiconductor Devices · explanation_only $\sigma = qn\mu_n + qp\mu_p$ sigma = q*n*mu_n + q*p*mu_p sci:equation:semiconductor:low-field-conductivity sha256:0a4901af68c18a20902cd760f5b5fa55e81c7651510c2094fc9ad790e3cf4766 Evidence: sci-evidence:semiconductor-devices:carrier-conductivity Locator: Chapter 2, 2.2 Drift, equation (2.2.14)	Spin Coating · explanation_only $h \propto \omega^{-1/2}$ h proportional to omega^(-1/2) sci:equation:spin-coating:drying-limited-power-law sha256:4d478fd6bf071a065d61974b648f2124c1ae1f2425c0da53be0233ccc3513470 Evidence: sci-evidence:spin-coating:microchemicals-spin-speed-direction Locator: Influence of the Attained Spin Speed

수식 판정 경계

결정론적 Rule에 정확히 결합된 3개 수식만 closed evaluator의 equal/product/quotient 계약 안에서 사용됩니다. Materials Science, Semiconductor Devices, Spin Coating 수식 3개는 explanation_only입니다.

vector, matrix, derivative, integral, summation, chemical reaction은 검토된 설명:표시로 보존할 수 있지만 현재 결정 권한으로 지원하지 않습니다. 범용 CAS 단순화나 자동 재배열을 판정에 사용하지 않습니다.

설명과 digest 계약

GroundedExplanation은 Knowledge-Rule-Equation-evaluator-Evidence identity/digest를 구조화해 묶습니다.

VerificationReport.report_digest는 renderer 전용 표시 필드를 제외한 과학적 payload를 식별하고, Markdown/PDF export_digest는 실제 렌더링 바이트를 별도로 식별합니다.

승인 대기

이 PDF의 FINAL은 구현 증빙 묶음의 완료만 뜻합니다. 사람 Admin 승인과 독립 sealed holdout은 대기 중이며 Science Knowledge Release는 NOT ACTIVE입니다.